

Race 13

Pythagoras' Theorem

Topics
using Pythagoras' Theorem to find the length of the hypotenuse
using Pythagoras' Theorem to find the length of a shorter side
solving word problems using Pythagoras' Theorem

TEACHER'S NOTES

This activity will enable students to practice the full range of skills relating to Pythagoras' Theorem: finding the hypotenuse, finding a shorter side, determining whether a triangle is right-angled or not, and worded problems to solve.

ANSWERS

Round 1	Round 2
8.94 m	6.62 cm
Round 3	Round 4
5.06 cm	11.49 m
Round 5	Round 6
15.09 m	12.81 km
Round 7	Round 8
No You may decide you require working to justify their decision, to avoid guessing.	Yes You may decide you require working to justify their decision, to avoid guessing.
Round 9	Round 10
1.73 m	42 inches

Where appropriate, answers are given to 2 decimal places.

Race 14

Right-angled Trigonometry

Topics
using right-angled trigonometry to find the length of a side
using right-angled trigonometry to find the size of an angle
solving word problems using right-angled trigonometry

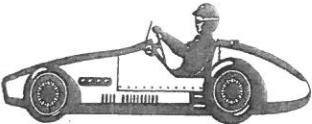
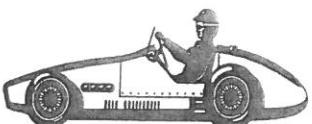
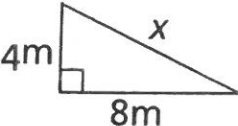
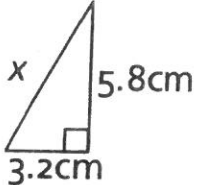
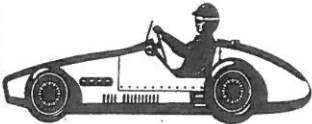
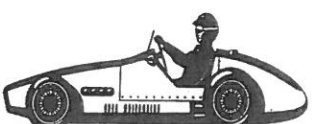
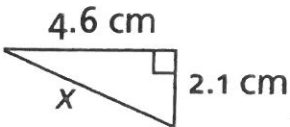
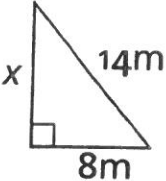
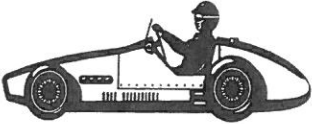
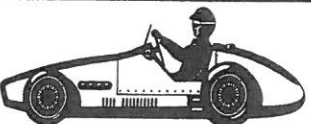
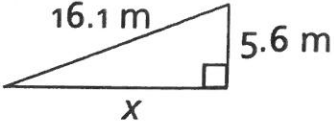
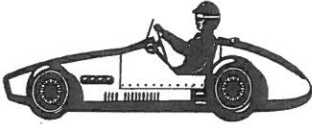
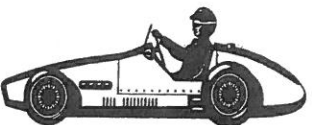
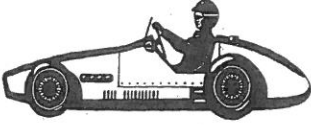
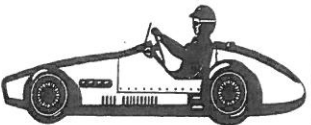
TEACHER'S NOTES

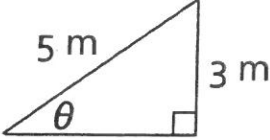
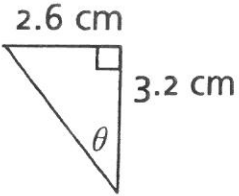
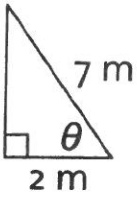
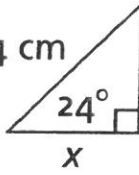
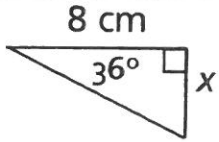
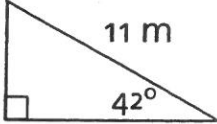
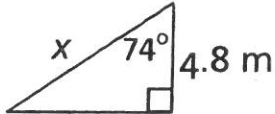
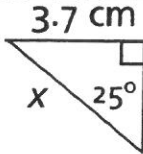
This team race contains questions covering all varieties of right-angled trigonometry questions. It includes use of the sine, cosine and tangent functions, and questions where students find shorter sides, hypotenuses and angles. There are also two worded questions in the last round. It might be worth pointing out to students that none of the diagrams are drawn to scale.

ANSWERS

Round 1	Round 2
$\theta = 36.9^\circ$	$\theta = 39.1^\circ$
Round 3	Round 4
$\theta = 73.4^\circ$	$x = 3.7 \text{ cm}$
Round 5	Round 6
$x = 5.8 \text{ cm}$	$x = 7.4 \text{ cm}$
Round 7	Round 8
$x = 17.4 \text{ m}$	$x = 8.8 \text{ cm}$
Round 9	Round 10
The foot of the ladder should be placed 0.7 metres from the wall.	31.0°

All answers are given correct to 1 decimal place.

	Round 1		Round 2
Find the length of side x		Find the length of side x	
	Round 3		Round 4
Find the length of side x		Find the length of side x	
	Round 5		Round 6
Find the length of side x		A man on a hike parks at a car park and walks 8 kilometres due East. He stops for half an hour and then travels due South for 10 kilometres. How far would he have to walk to the car park if he could walk directly there?	
	Round 7		Round 8
The two shorter sides of a triangle are 3.6 cm and 2.4 cm. The hypotenuse is 5.1 cm. Is the triangle right-angled?		The three sides of a triangle are 12.7cm, 8.2 cm and 15.1 cm, rounded to the nearest mm. Is the triangle right-angled?	
	Round 9		Round 10
Sam places a 2.5 m ladder against a wall, so that it reaches a height of 1.8 m up the wall from the ground. How far from the wall is the base of the ladder?		The height of a TV screen is 25.2 inches, and the width is 33.6 inches. What is the length of the diagonal of the TV screen?	

	Round 1		Round 2
Find the size of angle θ :		Find the size of angle θ :	
	Round 3		Round 4
Find the size of angle θ :		Find the length of side x :	
	Round 5		Round 6
Find the length of side x :		Find the length of side x :	
	Round 7		Round 8
Find length of side x :		Find the length of side x :	
	Round 9		Round 10
A man places a 2 metre ladder against a wall. The ladder is considered safe if the angle between the ladder and the floor is 70° . How far from the wall should the foot of the ladder be placed?		A vertical post of length 3 metres supports one end of a bridge. A suspension cable is attached to the top of the post and anchored to the ground 5 metres away. What angle does the cable make with the ground?	